



East West University

Department of Computer Science and Engineering

Semester : spring 2021

Course Title: Electrical Circuits

Course Code: CSE 209

Section: 03

Experiment No: 3

Experiment Title: Bias Point Detail Analysis of DC Circuit With Independent Sources Using PSpice Schematics.

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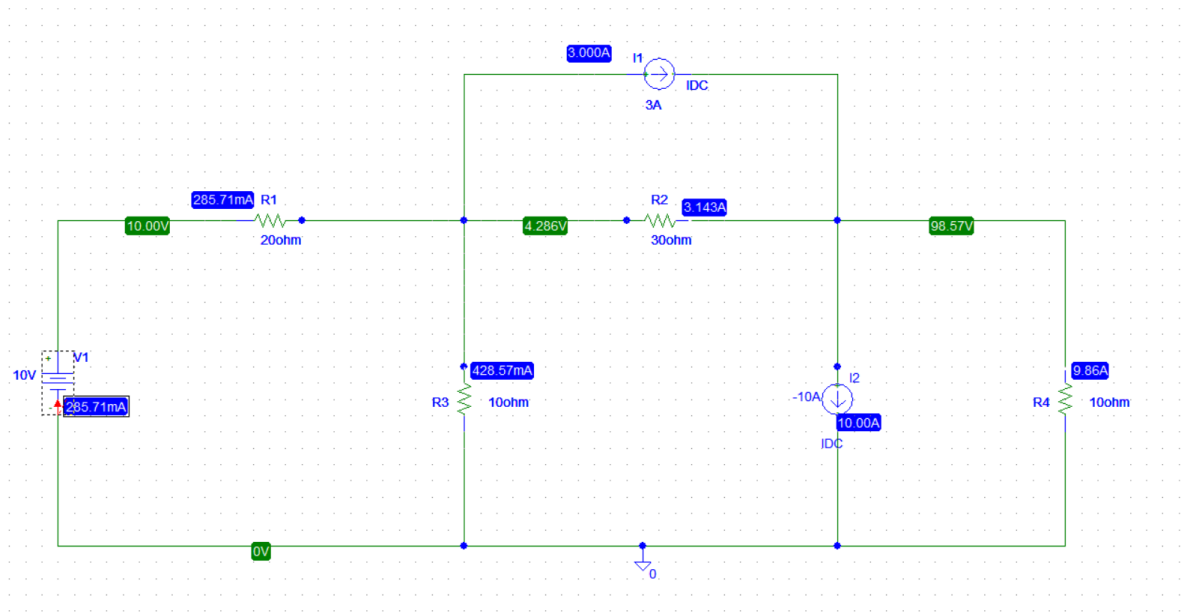
Date of Performance: 10-3-2021

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Objectives:

1. To learn fundamentals of PSpice.
2. To analyze Bias Point Detail of DC circuit using PSpice Schematics.

Circuit Diagram:



Answers to the post lab report questions:

1)

Applying KCL at node 1,

$$V_1 = 10 \text{ V}$$

Applying KCL at node 2,

$$\frac{V_2 - V_1}{20} + \frac{V_2}{10} + \frac{V_2 - V_3}{30} = -3$$

$$\frac{V_2 - 10}{20} + \frac{V_2}{10} + \frac{V_2 - V_3}{30} = -3$$

$$V_2 \left(\frac{1}{20} + \frac{1}{10} + \frac{1}{30} \right) + V_3 \left(-\frac{1}{30} \right) = -3 + \frac{10}{20}$$

$$.1833V_2 - .0333V_3 = -2.5 \dots\dots\dots (1)$$

Applying KCL at node 3,

$$\frac{V_3 - V_2}{30} + \frac{V_3}{10} = 3 + 10$$

$$V_2 \left(-\frac{1}{30}\right) + V_3 \left(\frac{1}{30} + \frac{1}{10}\right) = 13$$

$$-0.0333V_2 + 0.1333V_3 = 13 \dots\dots\dots (2)$$

Solving equation (1) and (2) we get,

$$V_2 = 4.272 \text{ V}$$

$$V_3 = 98.592 \text{ V}$$

Calculation of the current,

$$R_1, I_1 = \frac{V_1 - V_2}{20} = 0.2864 \text{ A} = 286.4 \text{ mA}$$

$$R_2, I_2 = \frac{V_3 - V_2}{30} = 3.144 \text{ A}$$

$$R_3, I_3 = \frac{V_2}{10} = 0.4272 \text{ A} = 427.2 \text{ mA}$$

$$R_4, I_4 = \frac{V_3}{10} = 9.8592 \text{ A}$$

2. Comparing the theoretically calculated values with the simulated values from PSpice.

	Theoretical value	Obtained value
Current	$I_1 = 286.4 \text{ mA}$	$I_1 = 285.71 \text{ mA}$
	$I_2 = 3.144 \text{ A}$	$I_2 = 3.143 \text{ A}$
	$I_3 = 427.2 \text{ mA}$	$I_3 = 428.57 \text{ mA}$
	$I_4 = 9.8592 \text{ A}$	$I_4 = 9.86 \text{ A}$
Voltage	$V_1 = 10 \text{ V}$	$V_1 = 10 \text{ V}$
	$V_2 = 4.272 \text{ V}$	$V_2 = 4.286 \text{ V}$
	$V_3 = 98.592 \text{ V}$	$V_3 = 98.57 \text{ V}$

The experimental values of the voltages and the currents are slightly different from the theoretical values.

Conclusion:

From this experiment we have learnt about analyzing bias point detail of a DC circuit, to calculate currents & voltages of a circuit using KCL. We have also learnt about a little discrepancy between the calculated & simulated values.